SHIVAM KUMAR

Embedded Systems Engineer — ECE Undergraduate

Email ◇ Phone◇ linkedin ◇ GitHub◇ Portfolio

PROFILE SUMMARY

Electronics & Communication Engineering student with a strong foundation in embedded systems, programming and digital hardware design. Hands-on experience in IoT, VLSI and firmware-based projects, with strong problem-solving skills and a focus on developing practical hardware-software solutions.

EDUCATION

Bachelor of Technology in Electronics & Communication Engineering — Imphal , Manipur
Indian Institute of Information Technology, Manipur. — 2023-2027

Higher Secondary Education (Class XII) — Madhubani, Bihar
P D K J College, Andhrathari, Madhubani (2020-2022) — 82.8

SKILLS

Programming Languages	C, C++, Python, HTML, CSS
Embedded Systems	ESP32, Arduino, Raspberry Pi, Zynq, Microcontrollers, Sensors & Actuators
Tools & Software	MATLAB, GNU Octave, Keil uVision, Tinkercad, Git, VS Code.
Core Concepts	Digital & Analog Electronics , Data Structures & Algorithms (C++), RTOS, UART, SPI, I2C, Signal Processing , Debugging , OS , OOP, DBMS.
Languages	English, Hindi, Maithili (Native)

EXPERIENCE

Skillbit Technologies | Embedded Systems + IoT Intern | Remote — **May 2026 – Jul 2026**

- Designed and developed an ESP32-based Home Automation System using relay modules to remotely control household appliances through an IoT platform.
- Built an IoT-based Gas Leakage Detection System integrating gas sensors with ESP32 to detect hazardous gas levels and generate real-time alerts
- Interfaced sensors, relays, and microcontrollers while implementing embedded programming and hardware debugging techniques.
- Gained practical experience in Embedded C/C++, ESP32, IoT, sensor interfacing, and embedded system development.

PROJECTS

ESP32 GPS Monitoring System: (Project Link)
Technologies: ESP32, GPS Module, SIM900A GSM, Embedded C/C++, Arduino IDE

- Developed a real-time GPS monitoring system using ESP32, GPS, and GSM modules for location tracking.
- Integrated GPS data acquisition with GSM-based communication to transmit location coordinates via SMS.
- Implemented embedded firmware for serial communication, GPS data parsing, and seamless integration of hardware modules, followed by system testing and debugging to ensure reliable performance.

Design and Implementation of 16-bit Carry Skip Adder (CSKA): (Project Link)
Technologies: Verilog HDL, Xilinx Vivado, Cadence Virtuoso

- Led a 4-member team in the design and implementation of a 16-bit Carry Skip Adder (CSKA), coordinating task allocation, progress tracking, and technical discussions.
- Designed and verified modular components including the Full Adder, 4-bit Ripple Carry Adder (RCA), and Carry Skip logic using Verilog HDL.
- Simulated the digital design in Xilinx Vivado and contributed to the transistor-level implementation and verification in Cadence Virtuoso.